

Frozen-energy relative Class-II structural reduction

TECT verification-first repository · claim A10-CLASSII-RELATIVE-COMMUTATOR-REDUCTION

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1. Purpose and scope

Proved here. The A9 frozen variable is fixed with the new-shell trace Γ_j , while its coefficient commutator uses the cumulative trace $\Gamma_{\leq j}$. The raw retained-energy identity is exact. Its sharp cost-free retained fraction is $\theta = 1$, even for legitimate hard dyadic splits. After covariance normal ordering there is nevertheless a strict common-phase plane-wave direction with zero frozen energy and a negative commutator. Therefore every cutoff-uniform relative theorem must spend a strictly positive entropy budget. We also prove the exact conditional composition theorem that turns such a relative estimate together with a quantitative action-recovery estimate into the A7 Nelson bound.

Open boundary. This note does not prove the relative form bound or its equivalent stabilised log-Laplace estimate, and it does not control the positive past-energy mismatch needed to reconstruct the actual A7 action. Those are the two named gates **A10-CLASSII-MULTISCALE-ACTION-DECOMPOSITION** and **A10-CLASSII-STABILISED-RELATIVE-LOG-LAPLACE**. A sharp rectangular-cube filtration subgate is closed here. The self-coupled Gibbs measure is not constructed.

The authority is the hash-pinned A1 production Gaussian on $\mathbb{T}_L^3$, $L = 16$, with three complex components, fixed $\rho_{\text{floor}} = 10^{-12}$, exact products, and the common real-even scalar regulator class already declared by A7–A9. The Class-II coefficient matrix is

$$B(X) = \sum_{A=1}^3 [p_A(X) \ v_A(X)] Q_{\text{II}} [p_A(X) \ v_A(X)]^T, \quad (1.1)$$

where

$$p_A = 2S_A X, \quad v_A = 2(S_A - q_A I)X, \quad q_A = \frac{X^T S_A X}{|X|^2 + \rho_{\text{floor}}}, \quad (1.2)$$

and

$$Q_{\text{II}} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} > 0, \quad (a, b, c) = (0.0045, 0.001875, 0.00234375) \quad (1.3)$$

up to the displayed hash-pinned mass-regulariser rounding.

1 The exact frozen-shell convention

Let

$$x = \phi_{j-1}, \quad \xi_j = \phi_j - \phi_{j-1}, \quad z = \phi_j, \quad (2.1)$$

and let

$$\Gamma_{j,i} = \mathbb{E}[D_i \xi_j D_i \xi_j^T], \quad \Gamma_{\leq j,i} = \sum_{\ell \leq j} \Gamma_{\ell,i}. \quad (2.2)$$

The A6/A7 convention is load-bearing here. If $D_i = \mathbb{E}[\partial_i \Psi \partial_i \Psi^\dagger]$, then $\mathbb{E}[\Psi_k \Psi_k^\dagger] = 2A(k)^{-1}$ and $\Gamma_i = \frac{1}{2} \text{realify}(D_i)$. Hence, after summing over directions,

$$\Gamma_{\leq j} = \text{realify} \left(\frac{1}{V} \sum_{k \in \Lambda_{\leq j}} |k|^2 A(k)^{-1} \right), \quad \Lambda_{\leq j} := \{k_n : \max_a |n_a| \leq N_j\}. \quad (2.2a)$$

There is no further factor one half in (2.2a). For a matrix field b define

$$q_b(w) = \frac{1}{2} \sum_i \int (D_i w)^T b(D_i w), \quad t_\Gamma(b) = \frac{1}{2} \sum_i \int \text{Tr}(\Gamma_i b). \quad (2.3)$$

The noncentral determinant in A9 integrates only the independent innovation ξ_j . Its exact covariance-normal variable is therefore

$$\boxed{Q_j^{\text{fr}} = q_{B(x)}(z) - t_{\Gamma_j}(B(x)).} \quad (2.4)$$

The trace in (2.4) is Γ_j , not $\Gamma_{\leq j}$. The coefficient increment is

$$\boxed{\mathcal{C}_j = q_{B(z)-B(x)}(z) - t_{\Gamma_{\leq j}}(B(z) - B(x)).} \quad (2.5)$$

Consequently, for $0 \leq \theta \leq 1$,

$$\begin{aligned} R_j^\theta := \theta Q_j^{\text{fr}} + \mathcal{C}_j &= q_{B(z)}(z) - (1 - \theta)q_{B(x)}(z) - t_{\Gamma_{\leq j}}(B(z)) \\ &\quad + t_{\Gamma_{\leq j-1}}(B(x)) + (1 - \theta)t_{\Gamma_j}(B(x)). \end{aligned} \quad (2.6)$$

This identity is bookkeeping, but it is load-bearing. Using a cumulative trace in (2.4) would define a different random variable and would invalidate the A9 determinant factor.

2 Production growth bounds

Positivity of Q_{II} makes (1.1) positive semidefinite. Since

$$|p_A| \leq 2|X|, \quad |v_A| \leq 4|X|, \quad |Dp_A[h]| \leq 2|h|, \quad (3.1)$$

and direct differentiation of (1.2) gives

$$|Dq_A(X)[h]| |X| \leq 4|h|, \quad |Dv_A(X)[h]| \leq 12|h|, \quad (3.2)$$

the three generators yield

$$\boxed{\|B(X)\|_F \leq \beta_B |X|^2, \quad \beta_B = 12a + 48|b| + 48c = 0.2564999999999359,} \quad (3.3)$$

$$\boxed{\|DB(X)[h]\|_F \leq \beta_1 |X| |h|, \quad \beta_1 = 24a + 192|b| + 288c = 1.1429999999997142.} \quad (3.4)$$

Integrating (3.4) along the segment from x to z gives

$$\|B(z) - B(x)\|_F \leq \frac{\beta_1}{2} (|x| + |z|) |z - x|. \quad (3.5)$$

These bounds are deliberately coarse and global; no derivative of the fixed positive floor is hidden in a constant.

3 Raw retained energy and its sharp dyadic threshold

Write

$$q_x = q_{B(x)}(z), \quad q_z = q_{B(z)}(z). \quad (4.1)$$

The raw part of (2.5) is $\mathcal{C}_j^{\text{raw}} = q_z - q_x$, and

$$\boxed{\theta Q_{j,\text{raw}}^{\text{fr}} + \mathcal{C}_j^{\text{raw}} = q_z - (1 - \theta)q_x.} \quad (4.2)$$

Thus $\mathcal{C}_j^{\text{raw}} \geq -q_x$, and $q_x = 0$ implies $\mathcal{C}_j^{\text{raw}} = q_z \geq 0$. The choice $\theta = 1$ leaves $q_z \geq 0$. We now prove that no smaller fixed θ is a cost-free all-field choice, even when x and z must arise from a genuine dyadic Fourier split.

3.1 Strict-dyadic Blaschke lemma

Fix $0 < a_0 < 1$ and the inner function

$$F(w) = \frac{w - a_0}{1 - a_0 w}, \quad |F(e^{it})| = 1. \quad (4.3)$$

Its analytic Fourier partial sum is

$$F_N = -a_0 + (1 - a_0^2) \sum_{n=1}^N a_0^{n-1} e^{int}. \quad (4.4)$$

Set

$$x_m = F_m e_1, \quad z_m = F_{2m} e_1. \quad (4.5)$$

Then $x_m = P_{\leq m} z_m$ and the innovation contains exactly the modes $m < n \leq 2m$. On this physical scalar ray the Class-II density with a base f and derivative carrier d is

$$e_B(f; d) = \frac{1}{2} \left[a + 2bs_f + cs_f^2 \right] \left(2 \operatorname{Re}(\bar{f}, d) \right)^2, \quad s_f = \frac{\rho_{\text{floor}}}{|f|^2 + \rho_{\text{floor}}}. \quad (4.6)$$

The fields F_N converge uniformly to the unit-modulus F , so the bracket in (4.6) stays between two positive constants.

An exact autocorrelation calculation gives

$$|F_N|^2 = 1 - (1 - a_0^2)a_0^{2N} - 2(1 - a_0^2) \sum_{k=1}^N a_0^{2N-k} \cos(kt), \quad (4.7)$$

hence

$$\|\partial_t |F_{2m}|^2\|_2^2 = 2(1 - a_0^2)^2 \sum_{k=1}^{2m} k^2 a_0^{8m-2k} = O_{a_0}(m^2 a_0^{4m}). \quad (4.8)$$

This bounds $q_z = O_{a_0}(m^2 a_0^{4m})$. On the other hand, the sine mode $m + 1$ of

$$J_{\text{fr},m} = 2 \operatorname{Re}(\overline{F_m} \partial_t F_{2m}) \quad (4.9)$$

has coefficient

$$-2 \sum_{p=0}^{m-1} (p + m + 1) c_p c_{p+m+1} = -2a_0^{m+1} (1 + o(1)), \quad (4.10)$$

where $c_0 = -a_0$ and $c_n = (1 - a_0^2)a_0^{n-1}$. Parseval therefore yields $q_x \geq c_{a_0} a_0^{2m}$ for all large m . Combining (4.8)–(4.10),

$$\frac{q_z}{q_x} = O_{a_0}(m^2 a_0^{2m}) \longrightarrow 0, \quad \frac{-\mathcal{C}_j^{\text{raw}}}{q_x} \longrightarrow 1. \quad (4.11)$$

For every $\theta < 1$, (4.2) is negative along (4.5) for all sufficiently large m . Thus

$$\boxed{\theta_{\text{raw}, \text{cost-free}} = 1} \quad (4.12)$$

is sharp under strict dyadic geometry. The executable at $a_0 = 1/2$ reaches 0.999899249067 by $m = 10$ and separately verifies (4.7)–(4.8); this number illustrates the theorem and is not its proof.

4 Covariance trace: an exact zero-frozen negative direction

Let k lie strictly inside one new shell and choose

$$x = 0, \quad z(r) = Ae^{ik \cdot r} e_1. \quad (5.1)$$

Then $B(x) = 0$, so $Q_j^{\text{fr}} = 0$ with either the raw or the correct A9 trace convention. Every density ρ and moment m_A is constant. Therefore

$$J_A = \nabla m_A = 0, \quad K_A = J_A - \frac{m_A}{\rho + \rho_{\text{floor}}} \nabla \rho = 0, \quad (5.2)$$

and $q_{B(z)}(z) = 0$ exactly. Nevertheless $B(z)$ is nonzero PSD and the cumulative derivative covariance is positive definite. Hence

$$\boxed{C_j = -t_{\Gamma \leq j}(B(z)) < 0, \quad Q_j^{\text{fr}} = 0.} \quad (5.3)$$

No θ , including $\theta = 1$, makes the complete covariance-normal R_j^θ algebraically nonnegative. The former gate wording requiring this direction to be excluded is therefore false. The correct task is to classify and entropy-control every such direction.

For the sharp cube the A6/A7 contraction theorem gives

$$\frac{C_j}{L^3 N} \longrightarrow -\delta_{\text{cube}} W_\varepsilon(Ae_1), \quad \delta_{\text{cube}} = 0.01619898645075695, \quad (5.4)$$

where $s_{\text{dbl}} = |\Psi_1|^2 + |\Psi_2|^2$ is the active-doublet density and $\rho = |\Psi_1|^2 + |\Psi_2|^2 + |\Psi_3|^2$,

$$W_\varepsilon = 9(a + 2b + c)s_{\text{dbl}} - \frac{6bs_{\text{dbl}}^2}{\rho + \rho_{\text{floor}}} - \frac{3cs_{\text{dbl}}^2(\rho + 2\rho_{\text{floor}})}{(\rho + \rho_{\text{floor}})^2}. \quad (5.5)$$

For Ae_1 , $W_\varepsilon > 0$, and

$$\frac{W_\varepsilon(Ae_1)}{A^2} \longrightarrow 9a + 12b + 6c = 0.07706249999998072. \quad (5.6)$$

The large-amplitude trace-loss coefficient is therefore $0.0012483343933611451L^3NA^2$.

5 Why a positive entropy budget is necessary

Consider a proposed uniform relative estimate

$$\mathbb{E}_\nu \sum_{j \leq J} R_j^\theta \geq -\alpha_c \mathcal{H}(\nu \mid \gamma_J) - \epsilon_6 \mathbb{E}_\nu \|\phi_J\|_6^6 - K_\theta \mathbb{E}_\nu \|\phi_J\|_4^4 - C_\theta. \quad (6.1)$$

If $\alpha_c = 0$, take the plane-wave mean in (5.1) at the top shell and the finite-dimensional absolutely continuous law

$$\nu_{N,\sigma} = N(h_N, \sigma^2 C_N). \quad (6.2)$$

At each fixed N , let $\sigma \downarrow 0$. The fixed positive floor makes all displayed finite-dimensional functions continuous with polynomial growth. Equation (5.4) sends the left side to $-dNA^2$, while the quartic and sextic terms are $O(A^4)$ and $O(A^6)$ independently of N . No uniform C_θ can repair (6.1). Thus

$$\boxed{\alpha_c > 0 \text{ is necessary.}} \quad (6.3)$$

This defect is not a counterexample when entropy is present. If $\sum_i \|\Gamma_{\leq j, i}\|_F \leq C_\Gamma K$ and the production symbol obeys

$$A(k) \geq c_{\text{sym}}(1 + |k|^2)^2 I, \quad c_{\text{sym}} = 0.10836453058711468, \quad (6.4)$$

then its trace loss is $O(KA^2)$, whereas the Cameron–Martin entropy of the same shift is at least $O(c_{\text{sym}}K^4A^2)$. Their ratio is $O(K^{-3})$. This proves entropy mitigation only for the identified plane-wave family; it does not prove the relative form bound for arbitrary adapted laws.

6 The registered triad budget is necessary, not global

For the A9 triad

$$g_K = \cos(Kx) + \cos(Ky) - \cos(K(x+y)), \quad (7.1)$$

exact Fourier convolution gives

$$\langle |\nabla g_K|^2 \rangle = 2K^2, \quad \langle g_K |\nabla g_K|^2 \rangle = -K^2, \quad \langle g_K^2 |\nabla g_K|^2 \rangle = \frac{5}{2}K^2. \quad (7.2)$$

For $x = Ae_1$, $z = A(1 + \epsilon g_K)e_1$, and $A = tK$, define

$$c_F = 4a\epsilon^2, \quad c_C = a\epsilon^3(4 - 5\epsilon), \quad c_H = \frac{3}{2}Y\epsilon^2, \quad c_6 = \langle (1 + \epsilon g_K)^6 \rangle. \quad (7.3)$$

Optimising the leading polynomial in t^2 proves the exact necessary condition

$$\boxed{\alpha_c \epsilon_6 \geq \frac{[(c_C - \theta c_F)_+]^2}{4c_H c_6}.} \quad (7.4)$$

The no-extra-budget threshold on this family is

$$\theta \geq \epsilon - \frac{5}{4}\epsilon^2. \quad (7.5)$$

It is 3/16 at $\epsilon = 0.3$ and has supremum 7/36 over $0 < \epsilon < 1/3$. The Blaschke theorem (4.12) proves that neither number is a global raw threshold.

7 Exact action mismatch and the two remaining gates

Let the actual final-cutoff A7 energy at scale j be

$$V_j = q_{B(\phi_j)}(D\phi_j) - t_{\Gamma_{\leq j}}(B(\phi_j)). \quad (8.1)$$

The definitions (2.4)–(2.5), with $x = \phi_{j-1}$ and $z = \phi_j$, give the exact identity

$$\boxed{Q_j^{\text{fr}} + \mathcal{C}_j = V_j - V_{j-1} + q_{B(\phi_{j-1})}(D\phi_{j-1}).} \quad (8.2)$$

Indeed $Q_j^{\text{fr}} + \mathcal{C}_j = V_j + t_{\Gamma_{\leq j-1}}(B(x))$ and $t_{\Gamma_{\leq j-1}}(B(x)) = q_{B(x)}(Dx) - V_{j-1}$. Consequently, when $V_0 = 0$,

$$\sum_{j \leq J} (Q_j^{\text{fr}} + \mathcal{C}_j) = V_J + E_J, \quad E_J := \sum_{j \leq J} q_{B(\phi_{j-1})}(D\phi_{j-1}) \geq 0. \quad (8.3)$$

This is an obstruction, not a favourable remainder: reconstructing V_J requires subtracting E_J . A lower bound for the larger shell expression does not lower-bound the actual A7 action. The gate **A10-CLASSII-MULTISCALE-ACTION-DECOMPOSITION** may be closed either by a true-increment determinant theorem or by a cutoff-uniform upper form bound

$$\mathbb{E}_\nu E_J \leq \alpha_d \mathcal{H}(\nu \mid \gamma_J) + \epsilon_d \mathbb{E}_\nu \|\phi_J\|_6^6 + K_d \mathbb{E}_\nu \|\phi_J\|_4^4 + C_d. \quad (8.4)$$

No such estimate is proved here.

For $\alpha_c > 0$ put

$$G_J = \sum_{j \leq J} R_j^\theta + \epsilon_6 \|\phi_J\|_6^6 + K_\theta \|\phi_J\|_4^4. \quad (8.5)$$

At finite cutoff, the Gibbs variational formula makes

$$\mathbb{E}_\nu G_J \geq -\alpha_c \mathcal{H}(\nu \mid \gamma_J) - C_\theta \quad (\nu \ll \gamma_J) \quad (8.6)$$

equivalent to

$$\boxed{\log \mathbb{E}_{\gamma_J} \exp(-G_J/\alpha_c) \leq C_\theta/\alpha_c.} \quad (8.7)$$

The uniform version is the separate gate **A10-CLASSII-STABILISED-RELATIVE-LOG-LAPLACE**. It controls the shell expression in (8.5), but (8.3) proves that it does not recover the actual A7 action by itself.

The A9 determinant alone cannot prove (8.7) by discarding endpoint terms. For every $\theta < 1$, $B(z) - (1-\theta)B(x)$ may be indefinite and the past x is unbounded. A successful proof must retain the self-coupled endpoint square or allocate the final sextic through a trace-safe mechanism.

8 Closed cube filtration and conditional composition

The filtration prerequisite can be closed for one concrete route. Put $N_j = 2^j$ and

$$P_{\leq j} = \mathbf{1}_{\{\max_a |n_a| \leq N_j\}}, \quad \xi_j = (P_{\leq j} - P_{\leq j-1})\phi. \quad (9.1)$$

The production Gaussian is diagonal in Fourier momentum, so disjoint blocks ξ_j are independent. In one dimension the symmetric partial-sum operator satisfies

$$S_N f = e^{-iNx} P_+(e^{iNx} f) - e^{i(N+1)x} P_+(e^{-i(N+1)x} f). \quad (9.2)$$

The M. Riesz theorem bounds P_+ on L^4 uniformly; Fubini and tensorisation therefore give $\|P_{\leq j}\|_{L^4(\mathbb{T}^3) \rightarrow L^4(\mathbb{T}^3)} \leq C_4^3$. Thus $\|\phi_{j-1}\|_4 \leq C_4^3 \|\phi_J\|_4$. Cube shells also have $O(2^{3j})$ modes of size $|k| \asymp 2^j$, so the A9 counting gives

$$\|T_{j,B(\phi_{j-1})}\|_{\mathfrak{S}_2}^2 \leq C_{\text{sh}}(L) M_R^4 c_{\text{sym}}^{-2} w_j \|B(\phi_{j-1})\|_2^2, \quad \sum_j w_j < \infty. \quad (9.3)$$

This proves the scoped subgate **A10-CLASSII-DYADIC-FILTRATION-REALISATION**. It does not extend to overlapping smooth Littlewood–Paley increments without another proof.

Write the actual production action as $S_J = U_J + V_J$, with the non-Class-II part satisfying

$$U_J \geq \int \left(\frac{\lambda}{4} \rho^2 + \frac{\gamma}{6} \rho^3 \right). \quad (9.3a)$$

Let $s = 1 - \theta$. The A9 noncentral determinant and conditional entropy chain rule yield

$$\mathbb{E}_\nu \sum_j s Q_j^{\text{fr}} \geq -\alpha_f \mathcal{H}(\nu \mid \gamma_J) - K_f \mathbb{E}_\nu \|\phi_J\|_4^4, \quad (9.4)$$

where

$$C_{\text{fr}} = C_{\text{sh}}(L)M_R^4 c_{\text{sym}}^{-2} \beta_B^2 C_4^{12} S_{\text{dy}}, \quad K_f = \frac{s^2 C_{\text{fr}}}{4\alpha_f}. \quad (9.5)$$

Assume both open estimates (8.4) and (8.6). With the production lower bound $U_J \geq \int [(\lambda/4)\rho^2 + (\gamma/6)\rho^3]$, set

$$\alpha = \alpha_f + \alpha_c + \alpha_d, \quad B_6 = \frac{\gamma}{6} - \epsilon_6 - \epsilon_d, \quad A_4 = \left[K_\theta + K_f + K_d - \frac{\lambda}{4} \right]_+. \quad (9.6)$$

Equations (8.3), (8.4), (8.6), and (9.4) then give

$$\mathbb{E}_\nu S_J \geq -\alpha \mathcal{H}(\nu \mid \gamma_J) + B_6 \mathbb{E}_\nu \|\phi_J\|_6^6 - A_4 \mathbb{E}_\nu \|\phi_J\|_4^4 - (C_\theta + C_d). \quad (9.7)$$

If $B_6 > 0$ and some $p > 1$ satisfies $p\alpha < 1$, the Gibbs variational formula and the exact maximum of $A_4 r^2 - B_6 r^3$ give

$$\log \mathbb{E}_{\gamma_J} e^{-p S_J} \leq p(C_\theta + C_d) + \frac{4pL^3 A_4^3}{27B_6^2}. \quad (9.8)$$

This is the A7 Nelson estimate conditional on both named open gates.

At the production point, the proof-friendly target

$$\theta = 0.90, \quad \alpha_f = 0.05, \quad \alpha_c = 0.80, \quad \alpha_d = 0.04, \quad \epsilon_6 = 0.25, \quad \epsilon_d = 0.01, \quad p = 1.10 \quad (9.9)$$

has $p\alpha = 0.979$, $B_6 = 0.01$, and $K_f = 0.05C_{\text{fr}}$. K_d and C_d remain symbolic proof targets. None of these targets proves (8.4) or (8.7).

9 Reproduction and evidence discipline

The primary executable derives every production number from A1/A7 authority, checks the exact trace and action-reconstruction identities, reconstructs the Blaschke and plane-wave families, and verifies the conditional composition algebra. The independent route imports neither primary code nor its numerical outputs; it rebuilds the Pauli currents, covariance, Fourier quadrature, entropy chain rule, and quartic–sextic optimiser. The one-command verifier rehashes all sources and requires agreement before accepting either child result.

10 Devil’s-advocate self-test

1. **UPHELD as a defect – the frozen trace was left ambiguous.** The determinant integrates only the new innovation, so (2.4) now pins Γ_j . A cumulative trace there is rejected by the verifier.
2. **UPHELD as a repaired numerical defect – the six-real covariance was halved twice.** A pre-freeze audit found that both executable routes used $\frac{1}{2} \text{realify}(\sum_k |k|^2 A(k)^{-1}/V)$. Equation (2.2a) restores the A6/A7 complex factor two. The finite plane-wave trace values double, while their sign and the analytic coefficient in (5.4)–(5.6) are unchanged. Both routes now test convergence toward that coefficient, so a repeated factor error fails closed.
3. **UPHELD – 3/16 is the global retained fraction.** The strict-dyadic Blaschke family proves the raw loss ratio tends to one. 3/16 belongs only to the registered $\epsilon = 0.3$ triad.
4. **UPHELD as false – zero frozen energy excludes negative commutators.** Equation (5.3) is an exact counterexample after trace subtraction. The gate is corrected from exclusion to entropy control.

5. **DISMISSED – the plane-wave defect disproves every entropy bound.** Its trace-to-Cameron–Martin ratio is $O(K^{-3})$. It proves $\alpha_c > 0$ is necessary, not that positive α_c fails.
6. **VALID with mitigation – an A9 determinant split closes the remaining theorem automatically.** For $\theta < 1$ its conditional coefficient is indefinite on unbounded past fields. The open proof must retain endpoint self-coupling or allocate the stabilisers through a proved trace-safe route.
7. **UPHELD as false – the shell sum reconstructs the actual A7 action plus a harmless nonnegative remainder.** Equation (8.3) has the wrong direction for that inference: V_J is the shell sum *minus* E_J . An upper form bound such as (8.4), or a new true-increment theorem, is a separate load-bearing gate.
8. **UPHELD as an overclaim – (9.8) constructs the Gibbs measure.** Equation (9.8) is conditional on both (8.4) and (8.7). No density convergence, partition limit, or tightness is claimed here.
9. **VALID with mitigation – smooth dyadic cutoffs supply both independence and L^4 stability.** They need not. Section 9 instead closes a specific sharp rectangular-cube route by independent Fourier blocks and the tensorised M. Riesz bound; overlapping smooth increments remain outside it.
10. **UPHELD as an overclaim – this is T5/T6 closure.** The honest tier is T4 for the proved structural reduction. T5 requires operator reproduction and cannot enlarge the theorem; T6/T7 are excluded.

11 Result footer

Claim: A10-CLASSII-RELATIVE-COMMUTATOR-REDUCTION
 Tier: T4 PROVED-STRUCTURAL-REDUCTION@FIXED-FLOOR-FINITE-CUTOFF
 Closed: exact Gamma_j convention; B/DB constants; strict-dyadic raw
 theta=1 sharpness; zero-frozen trace defect; alpha_c>0 necessity;
 triad budget; exact action-mismatch identity; sharp-cube filtration;
 conditional Nelson-composition implication with two antecedents.
 Open: A10-CLASSII-MULTISCALE-ACTION-DECOMPOSITION;
 A10-CLASSII-STABILISED-RELATIVE-LOG-LAPLACE.
 Not claimed: the relative form bound, A7 Nelson closure, an interacting Gibbs
 measure, floor removal, infinite volume, BCC, T6, or T7.
 Primary expected: 47/47 PASS.
 Independent expected: 34/34 PASS.
 Integrated expected: 101/101 PASS.

References

1. M. Boué and P. Dupuis, “A variational representation for certain functionals of Brownian motion,” *Annals of Probability* 26 (1998), 1641–1659.
2. N. Barashkov and M. Gubinelli, “A variational method for Φ_3^4 ,” arXiv:1805.10814.